

## METABOLISM OF DIETHYLSTILBESTROL BY RAT LIVER: A PRELIMINARY REPORT

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*Aerobic incubation of a mixture of E[1,1,1-D<sub>3</sub>]-3,4-bis(p-hydroxyphenyl)-hex-3-ene admixed with an approximately equimolar amount of unlabeled diethylstilbestrol and [2-<sup>14</sup>C]diethylstilbestrol with rat liver homogenates in the presence of NADPH yielded water-soluble metabolites as well as products more and less polar than the starting material. Addition of S-adenosyl-L-methionine increased the quantity of nonpolar metabolites. Incubation with rat liver microsomes yielded similar results. When polar metabolites from incubation with rat liver microsomes were incubated with catechol O-methyltransferase and S-adenosyl[methyl-<sup>3</sup>H]-L-methionine there was conversion to <sup>3</sup>H-labeled nonpolar products. Examination of reaction products by means of gas chromatography-mass spectrometry gave evidence for the formation of a dihydroxydiethylstilbestrol, a dihydroxydienestrol, and a monomethoxydiethylstilbestrol.*

### INTRODUCTION

Our interest in the metabolism of diethylstilbestrol [Fig. 1 (1)] was aroused by the observations of Herbst et al. (1971) implicating this nonsteroidal estrogen as a causative factor of vaginal cancer in the daughters of women who had received the hormone during pregnancy. It occurred to us that—as is the case with many aromatic carcinogens (Gelboin et al., 1972)—the parent compound might not be the ultimate carcinogenic agent, but the precursor of a carcinogenically active metabolite. A second motivation was that, at the time of the initiation of this study, diethylstilbestrol was still being employed in the fattening of cattle and that, although analytical methods for its detection and measurement in cattle feed, meat, and other agricultural products were available, the measurement of residual metabolites in such products might be equally or even more important.

The procedure that we elected to use was the twin ion technique that had been developed (Braselton et al., 1973) for the recognition of steroid hormone metabolites by gas chromatography-mass spectrometry, which

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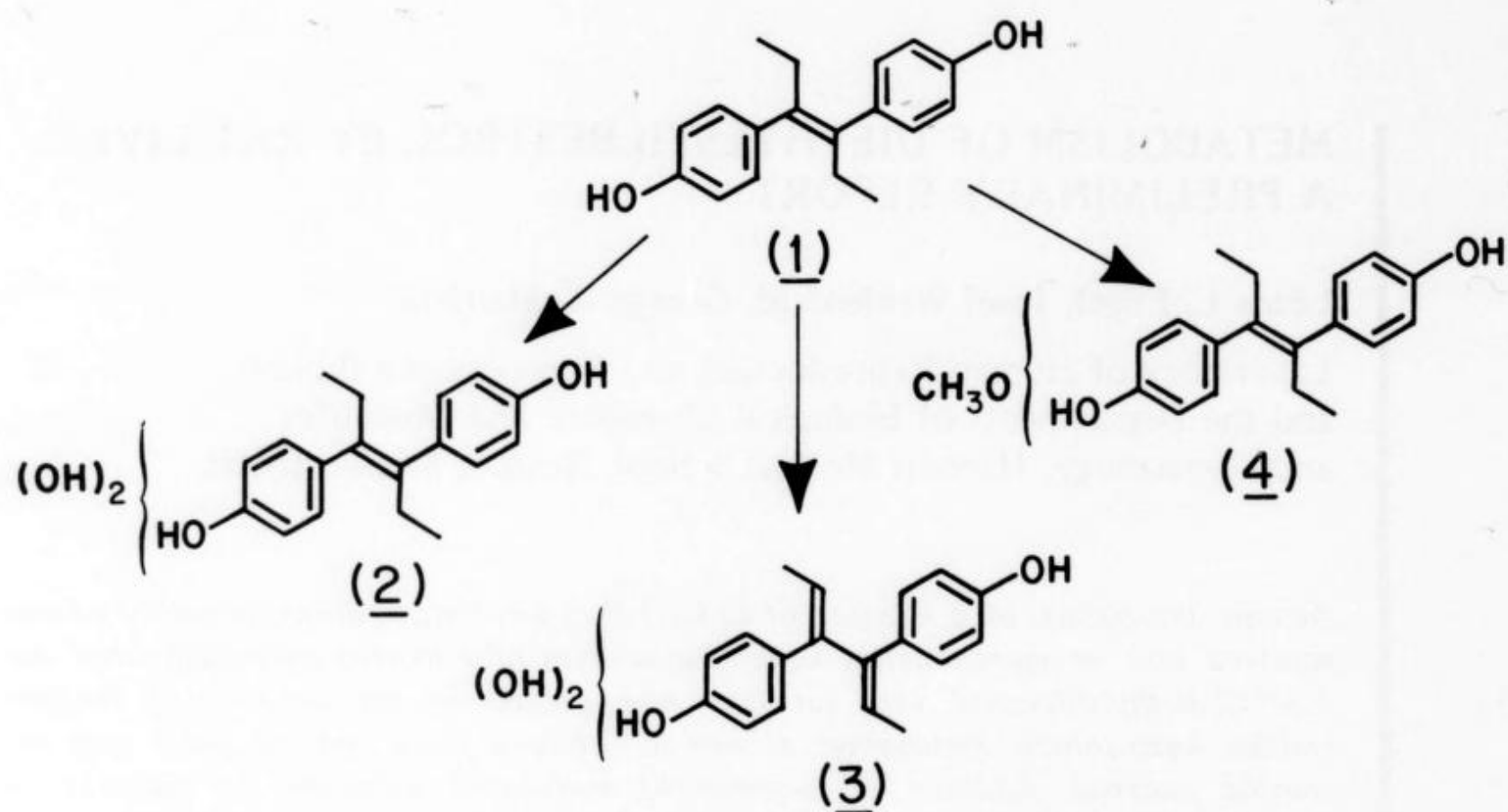


FIGURE 1. Metabolites of diethylstilbestrol (tentative).

depends on the availability of substrate labeled with a heavy isotope as well as with a radioisotope. Accordingly,  $[1,1,1-D_3]$  diethylstilbestrol was synthesized (Marshall and Engel, 1975). When this substrate is mixed with an approximately equimolar quantity of unlabeled diethylstilbestrol and examined in the gas chromatograph-mass spectrometer, ions at  $M^+$  and  $M+3$  are observed. Twin ions at  $M^+$  and  $M+3$  will also be found in metabolites that have not lost deuterium; indeed, appearance of ions at  $M+1$  and  $M+2$  in metabolites would provide evidence for reactions involving the terminal methyl groups of diethylstilbestrol. In our experiments a small amount of  $[^{14}C]$  diethylstilbestrol was used as a marker. Rat liver tissue was chosen for our preliminary studies. It was found that diethylstilbestrol is metabolized briskly both by rat liver homogenates and by liver microsomes. In this report we present some qualitative data on factors affecting the metabolism of diethylstilbestrol by rat liver preparations. The results indicate that such tissue extracts contain enzymes capable of metabolizing diethylstilbestrol.

## MATERIALS AND METHODS

### Animals

Female Sprague-Dawley rats, aged 60–80 days and weighing 220–280 g, were used. They were fed Purina Chow *ad libitum*. The rats were killed by cervical dislocation and the livers removed.

### Chemicals

All chemicals were analytical grade.

### Incubation Conditions

For the preparation of homogenates, 2 g liver was homogenized in 6 ml 200 mM sodium phosphate buffer, pH 7.2. For the preparation of microsomes, liver (2 g) was homogenized in 6 ml 0.25 M sucrose and subjected to differential centrifugation; the microsomal fraction was washed once with sodium phosphate buffer.

Unless indicated otherwise the substrate consisted of 40 nmol diethylstilbestrol, 40 nmol [1,1,1-D<sub>3</sub>]diethylstilbestrol (Marshall and Engel, 1975, and 0.2  $\mu$ Ci [<sup>14</sup>C]diethylstilbestrol (50 Ci/mol) (Amersham/Searle). The compounds in ethanol were added to the incubation flask and the solvent removed in a stream of nitrogen. S-Adenosyl[<sup>3</sup>H-methyl]-L-methionine had a specific activity of 12.3 Ci/mmol (New England Nuclear).

Incubations were performed in the buffer used for the preparation of homogenates. When microsomes were incubated, the buffer was also 10 mM in MgCl<sub>2</sub>. The total volume of the incubation mixture was 10 ml and the incubations were carried out at 37°C for 30 min.

Catechol O-methyl transferase was prepared from rat liver by the method of Coyle and Henry (1973).

### Processing the Extract

At the end of the incubation a mixture of acetone (50 ml) and methanol (50 ml), the latter containing 4% ascorbic acid and 1% acetic acid (Gelbke et al., 1973), was added and the precipitated protein centrifuged. The supernatant fraction was evaporated under reduced pressure, diluted to 60 ml with water, and extracted with two 60-ml portions of ether. The ether extract was evaporated to dryness, and the residue was partitioned between equal volumes (50 ml) of hexane and 70% aqueous methanol. The methanol was evaporated to dryness, and the residue partitioned between ethyl acetate (50 ml) and water (10 ml). The ethyl acetate was evaporated to dryness under reduced pressure, and the residue chromatographed on Whatman No. 1 paper in the system petroleum ether (bp 38.5–50.1°C) : toluene : methanol : water (100 : 100 : 85 : 15) (Bush, 1961). Three radioactive zones were detected in a Packard Radiochromatogram scanner: a polar zone, one corresponding in mobility to that of diethylstilbestrol, and a nonpolar zone. For further examination the radioactive zones were eluted with methanol and subjected to thin-layer chromatography on silica gel in the system hexane : ether : methylene chloride : ethanol (100 : 75 : 50 : 10) and examined by gas chromatography-mass spectrometry after conversion to trimethylsilyl ethers (Engel and Orr, 1972).

To retard oxidative destruction of metabolites, paper and thin-layer plates were impregnated with a solution of 4% ascorbic acid in methanol containing 1% acetic acid (v/v) (Gelbke et al., 1973).

### Gas Chromatography-Mass Spectrometry

An LKB 9000 instrument modified for solid sample introduction was used. The column was 1.75 m X 4 mm i.d., packed with 1% OV-22 on GasChrom Q and operated with a helium flow of 30 ml/min. Mass spectra were taken at 70 eV.

### RESULTS

When diethylstilbestrol was incubated with rat liver homogenates in the presence of NADPH, products more and less polar than the starting material were detected (Table 1). An appreciable amount of the radioactivity appeared in a water-soluble fraction and is presumed to be conjugated forms of diethylstilbestrol and its metabolites. This fraction has not been investigated further. Approximately equal quantities of metabolites more and less polar than diethylstilbestrol were detected by scanning of the paper chromatograms. When S-adenosyl-L-methionine was added to the incubation mixture, there was an increase in the quantity of radioactivity appearing in the nonpolar fraction and a corresponding decrease in the amount appearing in the polar fraction.

When diethylstilbestrol was incubated with rat liver microsomes, the distribution of radioactivity among the products was somewhat similar (Table 2). Under aerobic conditions in the presence of NADPH, an appreciable amount of radioactivity was found in the washed microsomal pellet, but the bulk of the radioactivity was divided between the water- and ether-soluble fraction; the ether-soluble fraction contained slightly more of the radioactivity. In the absence of NADPH or under anaerobic conditions, a negligible amount of radioactivity was found in the washed microsomal pellet; the distribution in the other two fractions was not affected.

Since variations in conditions had little effect on the amount of water-soluble radioactivity, attention was directed to the distribution of

TABLE 1. Metabolism of Diethylstilbestrol by Rat Liver Homogenates

| Cofactor                                    | Water soluble (nmol) | Ether soluble <sup>a</sup> (nmol) |                  |          |
|---------------------------------------------|----------------------|-----------------------------------|------------------|----------|
|                                             |                      | Polar                             | DES <sup>b</sup> | Nonpolar |
| NADPH (5 $\mu$ mol) (4)                     | 34 <sup>c</sup>      | 10                                | 11               | 9        |
| NADPH + SAM <sup>d</sup> (10 $\mu$ mol) (2) | 33                   | 6                                 | 12               | 17       |

<sup>a</sup> After paper chromatography.

<sup>b</sup> Diethylstilbestrol.

<sup>c</sup> All values are means. In this and other tables numbers in parentheses indicate numbers of experiments.

<sup>d</sup> S-Adenosyl-L-methionine.

TABLE 2. Metabolism of Diethylstilbestrol by Rat Liver Microsomes

| Incubation                | Microsomal pellet (nmol) | Water soluble (nmol) | Ether soluble (nmol) |
|---------------------------|--------------------------|----------------------|----------------------|
| NADPH (5 $\mu$ mol) (4)   | 8                        | 30                   | 40                   |
| No NADPH (3)              | <2                       | 30                   | 45                   |
| NADPH, N <sub>2</sub> (1) | <1                       | 35                   | 40                   |

radioactivity in the ether-soluble fraction after paper chromatography (Table 3). In the absence of NADPH or under anaerobic conditions, the bulk of the radioactivity had mobility similar to that of diethylstilbestrol. In the presence of NADPH and air, however, the radioactivity was divided approximately equally between polar products and those having mobilities similar to that of diethylstilbestrol. If the incubation mixture is fortified with *S*-adenosylmethionine, the radioactivity corresponding to metabolites is divided approximately equally between the polar and the nonpolar zones.

To confirm the involvement of *S*-adenosylmethionine in the metabolic transformations of diethylstilbestrol, an experiment was performed in which polar metabolites from several aerobic incubations of diethylstilbestrol with microsomes were combined and incubated with a partially purified preparation of catechol *O*-methyltransferase from rat liver (Coyle and Henry, 1973) and *S*-adenosyl-L-[methyl-<sup>3</sup>H]methionine. Table 4 shows the results of this experiment. With the complete system both <sup>3</sup>H and <sup>14</sup>C appear in the organic phase and cochromatograph on thin-layer chromatograms. In the absence of substrate negligible amounts of tritium appear in this fraction and, of course, there is no <sup>14</sup>C. When the polar fraction from an incubation with rat liver microsomes from which [<sup>14</sup>C]diethylstilbestrol was omitted is added to catechol *O*-methyltransferase

TABLE 3. Metabolism of Diethylstilbestrol by Rat Liver Microsomes—Effects of Cofactors on the Composition of the Ether-Soluble Fraction

| Incubation                              | Products <sup>a</sup> (nmol) |                  |          |
|-----------------------------------------|------------------------------|------------------|----------|
|                                         | Polar                        | DES <sup>b</sup> | Nonpolar |
| No cofactor, O <sub>2</sub> (3)         | 5                            | 30               | <2       |
| NADPH (5 $\mu$ mol), N <sub>2</sub> (1) | 5                            | 30               | <2       |
| NADPH, O <sub>2</sub> (6)               | 16                           | 15               | <2       |
| NADPH, O <sub>2</sub> + SAM (7)         | 8                            | 10               | 15       |

<sup>a</sup>After paper chromatography.

<sup>b</sup>Diethylstilbestrol.

TABLE 4. Incubation of Polar Metabolites of [ $^{14}\text{C}$ ] Diethylstilbestrol with Catechol *O*-Methyltransferase and *S*-Adenosyl-L-[methyl- $^3\text{H}$ ] methionine

| Incubation                                        | Incubation mixture <sup>a</sup><br>(dpm) |                   | Washed organic phase <sup>c</sup><br>(dpm) |                   | TLC peak zone<br>(dpm) |                   |
|---------------------------------------------------|------------------------------------------|-------------------|--------------------------------------------|-------------------|------------------------|-------------------|
|                                                   | $^3\text{H}$                             | $^{14}\text{C}$   | $^3\text{H}$                               | $^{14}\text{C}$   | $^3\text{H}$           | $^{14}\text{C}$   |
| Polar metabolites<br>(50 nmol), COMT <sup>b</sup> | $2.5 \times 10^6$                        | $2.8 \times 10^4$ | $10^6$                                     | $2.2 \times 10^4$ | $7.5 \times 10^5$      | $1.4 \times 10^4$ |
| COMT                                              | $2.5 \times 10^6$                        | 0.0               | $1.2 \times 10^4$                          | 0.0               | $10^3$                 | 0.0               |
| Polar fraction of<br>blank incubation             | $6 \times 10^5$                          | 0.0               | $3.6 \times 10^3$                          | 0.0               | —                      | —                 |
| DES (50 nmol)                                     | $2.5 \times 10^6$                        | 0.0               | $10^4$                                     | 0.0               | —                      | —                 |

<sup>a</sup>Tris·HCl, pH 7.8 (100 mM) (1.5 ml), 15  $\mu\text{mol}$   $\text{MgCl}_2$ , 1 mM ethylene glycol tetraacetic acid, 0.01% ascorbic acid.

<sup>b</sup>Catechol *O*-methyltransferase, prepared from rat liver.

and [ $^3\text{H}$ ]-*S*-adenosylmethionine or when catechol *O*-methyltransferase is omitted from an otherwise complete system, negligible amounts of tritium and no  $^{14}\text{C}$  appear on the thin-layer chromatogram. Finally, when diethylstilbestrol is incubated with catechol *O*-methyltransferase in the absence of the cofactor, neither tritium nor  $^{14}\text{C}$  appears in the nonpolar zone of the thin-layer chromatogram. Since no reference compounds were available, it was not possible to purify the products to constant specific activities.

Examination of samples of the polar and nonpolar fractions by gas chromatography-mass spectrometry was carried out after treatment with hexamethyldisilazane to form the trimethylsilyl ethers.

## CONCLUSIONS

At the time this work was begun, little was known about the metabolism of diethylstilbestrol by animal tissues save for formation of conjugates with glucuronic (Mazur and Shorr, 1942) or sulfuric acid (Starnes and Teague, 1949). No metabolites had been recognized in which more fundamental alterations of the molecule had been made. Gawienowski et al. (1962) reported the detection of an unconjugated metabolite of diethylstilbestrol in rat tissues after administration of the labeled compound.

The present results show that diethylstilbestrol is indeed attacked by enzymes present in rat hepatic microsomes. The requirements for air and reduced NADP strongly indicate that the initial attack is formation of an epoxide or introduction of a hydroxyl group, or both. That the hydroxyl group introduced into one or more of the phenolic rings is *ortho* to that already present is strongly suggested by the ability of catechol *O*-methyltransferase preparations to transfer methyl from *S*-adenosylmethionine to

TABLE 5. Metabolites of Diethylstilbestrol

| Compound                    | $M^+$ <sup>a</sup> | $R_c$ <sup>b</sup> |
|-----------------------------|--------------------|--------------------|
| Dihydroxydiethylstilbestrol | 588/591            | 0.3                |
| Dihydroxydienestrol         | 586/589            | 2.45               |
| Methoxydiethylstilbestrol   | 442/445            | 0.5                |

<sup>a</sup>The molecular ion  $M^+$  and the  $R_c$  values refer to the TMS derivatives.

<sup>b</sup>Retention time relative to 5 $\alpha$ -cholestane. Column temperature, 210°C; 1% OV-22 on GasChrom Q. Retention time for 5 $\alpha$ -cholestane is 14.6 min.

the mono- or dihydroxy derivatives of diethylstilbestrol. Since the enzyme preparation used in these experiments was not homogeneous, no firm conclusions can be drawn.

Examination of both polar and nonpolar metabolites formed by incubation of a mixture of deuterated and undeuterated diethylstilbestrol with rat hepatic microsomes gave evidence for the formation of a number of products containing the characteristic twin molecular ions at  $M^+$  and  $M+3$  (Table 5). The mass spectral evidence, which consists of molecular ions and a number of fragment ions, is consistent with the formation of a dihydroxy derivative of diethylstilbestrol (2) and a dihydroxy derivative of dienestrol (3). The latter compound could arise by attack on the stilbene double bond. It is not known which of the stereoisomers of dienestrol was formed. These compounds were obtained from the polar fraction. The nonpolar fraction from the incubation of the homogenate yielded material that gave a mass spectrum consistent with a monomethoxy derivative of diethylstilbestrol (4). Definitive identification of these compounds must await their isolation in pure form and the synthesis of reference compounds for direct comparison.

The results reported here are in substantial agreement with the findings recently published by Metzler (1975) who studied urinary and biliary metabolites of [1,1,1,2,2-D<sub>5</sub>]diethylstilbestrol administered to hamsters and rats. He found, in addition to unchanged diethylstilbestrol, a monohydroxy and a monomethoxy derivative of diethylstilbestrol, a dimethoxy derivative, dienestrol and the corresponding monohydroxy, and mono- and dimethoxy derivatives.

The experiments of Metzler taken in conjunction with our own observations clearly show that diethylstilbestrol is not biochemically inert. It is now a matter of considerable importance to identify definitively the metabolites already discovered and assay them for carcinogenic activity.

## DISCUSSION

W. E. HESTON: Would you care to comment on the bound radioactivity that you found in the microsomal precipitate?

L. L. ENGEL: I suspect that, as is true in the case of the catechol estrogens, the bound radioactivity is either a 2- or 4-thioether with sulfhydryl groups in some microsomal protein.

## REFERENCES

- Braselton, W. E., Jr., Orr, J. C. and Engel, L. L. 1973. The twin ion technique for detection of metabolites by gas chromatography-mass spectrometry: Intermediates in estrogen biosynthesis. *Anal. Biochem.* 53:64-85.
- Bush, I. E. 1961. *The chromatography of steroids*, pp. 45-59. Oxford: Pergamon Press.
- Coyle, J. T. and Henry, D. 1973. Catecholamines in fetal and newborn rat brain. *J. Neurochem.* 21: 61-67.
- Engel, L. L. and Orr, J. C. 1972. *Biochemical application of mass spectrometry*, ed. G. R. Waller, pp. 537-572. New York: Wiley-Interscience.
- Gawienowski, A. M., Knoche, H. W. and Moser, H. C. 1962. The metabolism of  $^{14}\text{C}$ - and  $^3\text{H}$ -labeled diethylstilbestrol in the rat. *Biochim. Biophys. Acta* 65:150-152.
- Gelbke, H. P., Kreth, M. and Knuppen, R. 1973. A chemical method for the quantitative determination of 2-hydroxysterone in human urine. *Steroids* 21:665-687.
- Gelboin, H. V., Kinoshita, N. and Wiebel, F. J. 1972. Microsomal hydroxylase: Induction and role in polycyclic hydrocarbon carcinogenesis and toxicity. *Fed. Proc.* 31:1298-1309.
- Herbst, A. L., Ulfelder, H. and Poskanzer, D. C. 1971. Adenocarcinoma of the vagina. *N. Engl. J. Med.* 284:878-881.
- Marshall, P. J. and Engel, L. L. 1975. The synthesis of trideuterated diethylstilbestrol ([1,1,1- $\text{D}_3$ ] *E*-3,4-di-(*p*-hydroxyphenyl)-hex-3-ene. *J. Labelled Comp.* 11:87-93.
- Mazur, A. and Shorr, E. 1942. The isolation of stilbestrol monoglycuronide from the urine of rabbits. *J. Biol. Chem.* 144:283-284.
- Metzler, M. 1975. Metabolic activation of diethylstilbestrol: Indirect evidence for the formation of a stilbene oxide intermediate in hamster and rat. *Biochem. Pharmacol.* 24: 1449-1453.
- Starnes, W. R. and Teague, R. S. 1949. Metabolism of synthetic estrogens. Urinary sulfur partitioning after estrogen administration. *J. Biol. Chem.* 179:43-47.